GATE ME 90 Days Challenge

Day 82/90 · Practice Set with Detailed Solutions

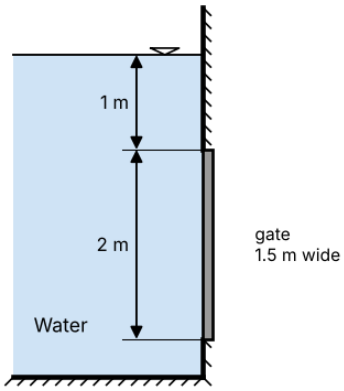
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Fluid & Thermal Sciences	Formula Sprint: Thermal	16 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a rapid formula revision of Fluid Mechanics (centre of pressure, Darcy–Weisbach, power transmission through pipes, venturimeter), Thermodynamics (Clausius inequality, irreversibility, wet-steam properties, isentropic relations), Heat Transfer (composite walls, lumped cooling, LMTD, radiation between concentric cylinders) and Applied Thermodynamics (centrifugal compressor work, feed-pump work, pump specific speed) — every solution states the key formula, so the set doubles as a revision sheet. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A vertical rectangular gate 1.5 m wide and 2 m high holds water on one side, as shown. Its top edge is 1 m below the free surface. The depth of the centre of pressure below the free surface is



(A) 2.000 m
(B) 2.333 m
(C) 2.167 m
(D) 2.250 m

Q2 NAT

Water flows at 1.8 m/s through a 250 mm diameter pipe 800 m long. The Darcy friction factor is 0.018. Taking g = 9.81 m/s², the head lost in friction (in m, round off to two decimal places) is _____

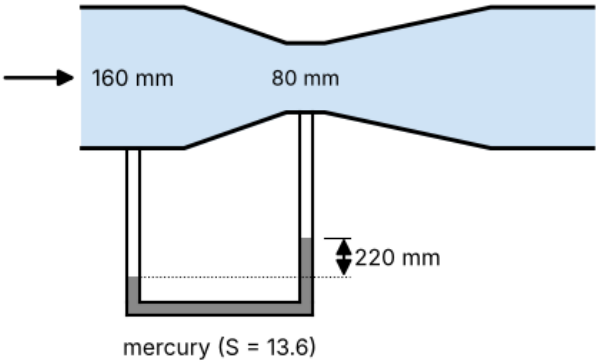
Q3 MCQ

Water is supplied to a hydraulic machine through a pipe 2 km long and 300 mm in diameter (Darcy f = 0.02) from a reservoir giving a total head of 300 m at the pipe inlet. Taking g = 9.81 m/s², the maximum power that can be delivered at the pipe outlet is about

(A) 532 kW
(B) 798 kW
(C) 266 kW
(D) 355 kW

Q4 NAT

A horizontal venturimeter has an inlet diameter of 160 mm and a throat diameter of 80 mm, as shown. A mercury (S = 13.6) differential manometer connected between them reads 220 mm. Taking C_d = 0.96 and g = 9.81 m/s², the discharge of water (in litres per second, round off to two decimal places) is _____



Q5 MCQ

An inventor claims that an engine receives 900 kJ of heat from a reservoir at 450 K, rejects 550 kJ to a reservoir at 300 K and produces 350 kJ of work per cycle. The claim is

(A) possible, and the engine is irreversible
(B) possible, and the engine is reversible
(C) impossible, because it violates the first law
(D) impossible, because it violates the Clausius inequality (second law)

Q6 NAT

A heat engine receives 1200 kJ of heat from a reservoir at 750 K and delivers 500 kJ of work. The surroundings are at 300 K. The irreversibility (lost work) of the engine, in kJ, is _____

Q7 MCQ

Wet steam at 0.2 MPa has a specific volume of 0.5 m³/kg. At 0.2 MPa: v_f = 0.001061 m³/kg, v_g = 0.8857 m³/kg, h_f = 504.7 kJ/kg and h_{fg} = 2201.6 kJ/kg. The specific enthalpy of the steam is about

(A) 1242 kJ/kg
(B) 1746 kJ/kg
(C) 1526 kJ/kg
(D) 2706 kJ/kg

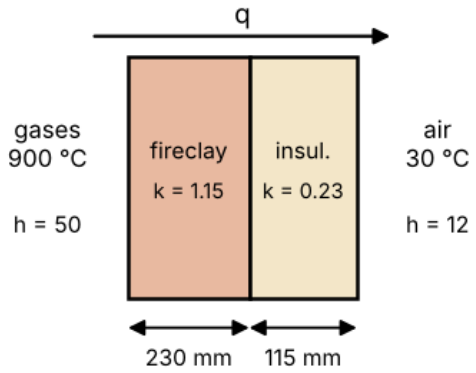
Q8 MCQ

Air (γ = 1.4) at 1 bar and 20 °C is compressed reversibly and adiabatically to 12 bar. Its final temperature is about

(A) 323 °C
(B) 41 °C
(C) 596 °C
(D) 247 °C

Q9 NAT

A furnace wall consists of 230 mm of fireclay brick (k = 1.15 W/m·K) and 115 mm of insulating brick (k = 0.23 W/m·K), as shown. The furnace gases are at 900 °C (h = 50 W/m²·K) and the outside air at 30 °C (h = 12 W/m²·K). The steady heat loss (in W/m², round off to one decimal place) is _____



Q10 NAT

A 4 mm thick aluminium plate (ρ = 2700 kg/m³, c = 900 J/kg·K, k = 205 W/m·K) at 300 °C is cooled from both faces by air at 25 °C with h = 40 W/m²·K. The time for the plate to cool to 100 °C (in s, round off to one decimal place) is _____

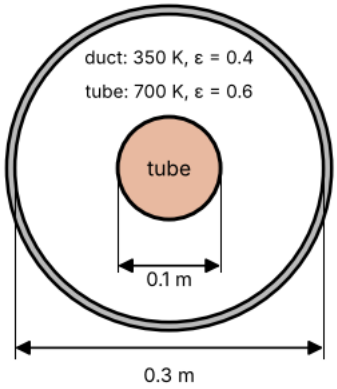
Q11 MCQ

In a heat exchanger the hot fluid cools from 140 °C to 80 °C and the cold fluid is heated from 25 °C to 65 °C. If it is arranged in counter flow, its log mean temperature difference is about

(A) 49.1 °C
(B) 55.0 °C
(C) 64.5 °C
(D) 65.0 °C

Q12 NAT

A long tube of 0.1 m outer diameter (surface temperature 700 K, ε = 0.6) runs concentrically inside a long duct of 0.3 m diameter (surface temperature 350 K, ε = 0.4), as shown. Taking σ = 5.67×10^{-8} W/m²·K⁴, the net radiation heat transfer per metre length (in W, round off to one decimal place) is _____



Q13 MCQ

A centrifugal compressor impeller of 0.5 m tip diameter runs at 15 000 rpm with radial vanes. The slip factor is 0.9 and the power input factor is 1.04. The work input per kg of air is about

(A) 154.2 kJ/kg
(B) 138.8 kJ/kg
(C) 160.4 kJ/kg
(D) 144.3 kJ/kg

Q14 MCQ

A boiler feed pump takes saturated water at 10 kPa (v = 0.00101 m³/kg) and delivers it to a boiler at 15 MPa. If the pump's isentropic efficiency is 80 %, the actual pump work is about

(A) 15.14 kJ/kg
(B) 18.92 kJ/kg
(C) 12.11 kJ/kg
(D) 0.19 kJ/kg

Q15 NAT

A centrifugal pump running at 1450 rpm delivers 0.08 m³/s against a head of 25 m. Its specific speed $N_S = N\sqrt{Q}/H^{3/4}$ (N in rpm, Q in m³/s, H in m) (round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(C) 2.167 m
Q2	NAT	9.51 (accept 9.46 to 9.56)
Q3	MCQ	(A) 532 kW
Q4	NAT	36.75 (accept 36.57 to 36.94)
Q5	MCQ	(D) impossible, because it violates the Clausius inequality (second law)
Q6	NAT	220
Q7	MCQ	(B) 1746 kJ/kg
Q8	MCQ	(A) 323 °C
Q9	NAT	1083.0 (accept 1077.6 to 1088.4)
Q10	NAT	157.9 (accept 157.1 to 158.7)
Q11	MCQ	(C) 64.5 °C
Q12	NAT	1850.6 (accept 1841.3 to 1859.8)
Q13	MCQ	(D) 144.3 kJ/kg
Q14	MCQ	(B) 18.92 kJ/kg
Q15	NAT	36.68 (accept 36.50 to 36.86)

Detailed Solutions

Q1 **Answer: (C) 2.167 m**

Concept: Vertical plane surface: $F = \rho g A\bar{h}$ and $h_{cp} = \bar{h} + I_G/(A\bar{h})$, with $I_G = bd^3/12$ for a rectangle.

$\bar{h} = 1 + 2/2 = 2$ m; $A = 1.5 \times 2 = 3$ m²; $I_G = 1.5 \times 2^3/12 = 1.0$ m⁴.

$h_{cp} = 2 + 1.0/(3 \times 2) =$ **2.167 m (C)**. (Force $F = 9810 \times 3 \times 2 = 58.9$ kN.)

Exam tip: $h_{cp} = 2d/3$ (2.333 m here) only when the top edge is AT the free surface. The centre of pressure is always below the centroid and approaches it as the gate goes deeper.

Q2 **Answer: 9.51 (accept 9.46 to 9.56)**

Concept: Darcy–Weisbach: $h_f = fLV^2/(2gD)$ with the Darcy f (= 4 × Fanning f). In terms of discharge, $h_f = 8fLQ^2/(\pi^2 g D^5)$.

$h_f = 0.018 \times 800 \times 1.8^2/(2 \times 9.81 \times 0.25)$.

$= 46.66/4.905 =$ **9.51 m**.

Exam tip: If a question gives the Fanning factor f′, use $h_f = 4f'LV^2/(2gD)$. Laminar flow: $f = 64/Re$. At constant Q, $h_f \propto 1/D^5$.

Q3 **Answer: (A) 532 kW**

Concept: Power transmitted $P = \rho g Q(H - h_f)$; it is maximum when $h_f = H/3$, giving a maximum transmission efficiency of 2/3.

$h_f = 100$ m: $V = \sqrt{(2gDh_f/(fL))} = \sqrt{(2 \times 9.81 \times 0.3 \times 100/(0.02 \times 2000))} = 3.836$ m/s.

$Q = (\pi/4)(0.3^2)(3.836) = 0.2712$ m³/s; $P = 9810 \times 0.2712 \times (300 - 100) = 532\,000$ W = **532 kW (A)**.

Exam tip: 798 kW uses the full 300 m (forgetting the friction loss). Efficiency of transmission $\eta = (H - h_f)/H$.

Q4 **Answer: 36.75 (accept 36.57 to 36.94)**

Concept: $Q = C_d A_1 A_2 \sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}$, where the piezometric head difference $h = x(S_m/S - 1)$ for a manometer reading x.

$h = 0.22 \times (13.6 - 1) = 2.772$ m of water; $\sqrt{(2gh)} = 7.375$ m/s.

$A_1 = 0.020106$ m², $A_2 = 0.005027$ m²; $A_1A_2/\sqrt{(A_1^2 - A_2^2)} = 0.005191$ m².

$Q = 0.96 \times 0.005191 \times 7.375 = 0.03675$ m³/s = **36.75 L/s**.

Exam tip: Using x × 13.6 instead of x × 12.6 is the classic slip. For a vertical or inclined venturimeter the same formula holds, because the manometer reads the piezometric head difference directly.

Q5 **Answer: (D) impossible, because it violates the Clausius inequality (second law)**

Concept: Clausius inequality: $\oint \delta Q/T \le 0$ (= 0 reversible, < 0 irreversible, > 0 impossible). Equivalent check: $W \le W_{Carnot} = Q_1(1 - T_2/T_1)$.

First law: 900 − 550 = 350 kJ, so energy is balanced.

$\oint \delta Q/T = 900/450 - 550/300 = 2.000 - 1.833 = +0.167$ kJ/K > 0.

Carnot limit = 900 × (1 − 300/450) = 300 kJ < 350 kJ → **impossible: violates the Clausius inequality (D)**.

Exam tip: Always take heat rejected as negative in $\oint \delta Q/T$. Satisfying the first law never makes a cycle feasible on its own.

Q6 **Answer: 220**

Concept: Maximum (reversible) work $W_{max} = Q(1 - T_0/T)$; irreversibility $I = W_{max} - W_{actual} = T_0 S_{gen}$; second-law efficiency $\eta_{II} = W/W_{max}$.

$W_{max} = 1200 \times (1 - 300/750) = 720$ kJ.

I = 720 − 500 = **220 kJ**; $\eta_{II} = 500/720 = 69.4$ % (while $\eta_I = 500/1200 = 41.7$ %).

Exam tip: Check with entropy: heat rejected = 700 kJ, $S_{gen} = 700/300 - 1200/750 = 0.7333$ kJ/K, and $T_0 S_{gen} = 220$ kJ (Gouy–Stodola).

Q7 **Answer: (B) 1746 kJ/kg**

Concept: Wet steam (quality x): $v = v_f + x\, v_{fg}$, $h = h_f + x\, h_{fg}$, $s = s_f + x\, s_{fg}$.

$x = (0.5 - 0.001061)/(0.8857 - 0.001061) = 0.564$.

$h = 504.7 + 0.564 \times 2201.6 =$ **1746 kJ/kg (B)**.

Exam tip: 1242 kJ/kg = $x\, h_{fg}$ forgets h_f. $x \approx v/v_g$ is fine for volume only when v_f is negligible — never write $h = x\, h_g$.

Q8 **Answer: (A) 323 °C**

Concept: Reversible adiabatic (isentropic) ideal gas: $T_2/T_1 = (p_2/p_1)^{(\gamma-1)/\gamma} = (V_1/V_2)^{\gamma-1}$, with T in kelvin.

$T_2 = 293.15 \times 12^{0.2857} = 293.15 \times 2.034 = 596.2$ K.

$T_2 = 596.2 - 273.15 =$ **323 °C (A)**.

Exam tip: Applying the ratio to 20 °C gives 41 °C, and 596 is the answer in K, not °C. With $pv^{1.3}$ = C the result would be 247 °C.

Q9 **Answer: 1083.0 (accept 1077.6 to 1088.4)**

Concept: Resistances in series per unit area: $R = 1/h_i + \Sigma L/k + 1/h_o$; $q = \Delta T_{overall}/R$ (electrical analogy).

$R = 1/50 + 0.23/1.15 + 0.115/0.23 + 1/12 = 0.020 + 0.200 + 0.500 + 0.0833 = 0.8033$ m²·K/W.

$q = (900 - 30)/0.8033 =$ **1083.0 W/m²**.

Surface temperatures: inside $900 - 1083/50 = 878.3$ °C; outside $30 + 1083/12 = 120.2$ °C.

Exam tip: Each layer's temperature drop is proportional to its resistance — the insulating brick (largest R) carries the largest drop. For a cylinder use $\ln(r_2/r_1)/(2\pi kL)$ instead of $L/(kA)$.

Q10 **Answer: 157.9 (accept 157.1 to 158.7)**

Concept: Lumped capacitance (valid for $Bi = hL_c/k < 0.1$): $(T - T_\infty)/(T_i - T_\infty) = e^{-t/\tau}$, with $\tau = \rho c L_c/h$ and $L_c = V/A$.

Plate cooled on both faces: L_c = half-thickness = 0.002 m; $Bi = 40 \times 0.002/205 = 0.00039$ → lumped OK.

$\tau = 2700 \times 900 \times 0.002/40 = 121.5$ s.

$t = \tau \ln[(300 - 25)/(100 - 25)] = 121.5 \times \ln 3.667 =$ **157.9 s**.

Exam tip: Using the full thickness as L_c doubles the time. L_c: plate (both faces) = t/2, long cylinder = r/2, sphere = r/3.

Q11 **Answer: (C) 64.5 °C**

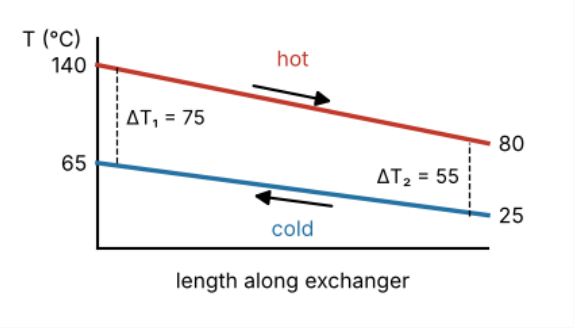
Concept: LMTD = $(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$; counter flow: $\Delta T_1 = T_{h,in} - T_{c,out}$, $\Delta T_2 = T_{h,out} - T_{c,in}$.

Counter flow: $\Delta T_1 = 140 - 65 = 75$ °C, $\Delta T_2 = 80 - 25 = 55$ °C.

LMTD = 20/ln(75/55) = **64.5 °C (C)**.

Parallel flow for the same terminal temperatures: ΔT = 115 and 15 °C → LMTD = 49.1 °C, so counter flow needs less area.

Exam tip: 65.0 °C is the arithmetic mean — always slightly larger than the LMTD. When $\Delta T_1 = \Delta T_2$, LMTD = ΔT.



Q12 **Answer: 1850.6 (accept 1841.3 to 1859.8)**

Concept: Two-surface grey enclosure: $Q_{12} = \sigma A_1(T_1^4 - T_2^4)/[1/\varepsilon_1 + (A_1/A_2)(1/\varepsilon_2 - 1)]$, since $F_{12} = 1$ for the inner surface.

Per metre: $A_1 = \pi \times 0.1 = 0.3142$ m²; $A_1/A_2 = 0.1/0.3 = 1/3$.

$\sigma A_1(T_1^4 - T_2^4) = 5.67 \times 10^{-8} \times 0.3142 \times (2.401 \times 10^{11} - 1.501 \times 10^{10}) = 4009.6$ W.

Denominator = 1/0.6 + (1/3)(1/0.4 − 1) = 1.6667 + 0.5 = 2.1667; Q = **1850.6 W per m**.

Exam tip: Using the parallel-plate expression ($A_1/A_2 = 1$) gives 1266 W. If the inner body is very small ($A_1/A_2 \to 0$), $Q = \varepsilon_1 \sigma A_1(T_1^4 - T_2^4)$.

Q13 **Answer: (D) 144.3 kJ/kg**

Concept: Centrifugal compressor (radial vanes, no inlet whirl): $W = \psi \sigma u_2^2$, where σ = slip factor, ψ = power input factor and $u_2 = \pi D_2 N/60$; stagnation temperature rise $\Delta T_0 = W/c_p$.

$u_2 = \pi \times 0.5 \times 15\,000/60 = 392.7$ m/s.

$W = 1.04 \times 0.9 \times 392.7^2 = 144\,340$ J/kg = **144.3 kJ/kg (D)** ($\Delta T_0 \approx 144/1.005 = 143.6$ K).

Exam tip: u_2^2 alone (154.2) ignores slip; σu_2^2 (138.8) is the energy given to the air before adding windage and disc-friction losses through ψ.

Q14 **Answer: (B) 18.92 kJ/kg**

Concept: Pump work for a liquid: $w_{p,ideal} = v\Delta p$ (v in m³/kg, Δp in kPa → kJ/kg); $w_{p,actual} = v\Delta p/\eta_p$.

$w_{p,ideal} = 0.00101 \times (15\,000 - 10) = 15.14$ kJ/kg.

$w_{p,actual} = 15.14/0.8 =$ **18.92 kJ/kg (B)**.

Exam tip: Divide by efficiency for a pump or compressor; multiply for a turbine (12.11 is the wrong direction). Δp in bar gives an answer 100 times too small (0.19).

Q15 **Answer: 36.68 (accept 36.50 to 36.86)**

Concept: Pump specific speed $N_S = N\sqrt{Q}/H^{3/4}$; turbine specific speed $N_S = N\sqrt{P}/H^{5/4}$. Affinity laws: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$.

$\sqrt{Q} = \sqrt{0.08} = 0.2828$; $H^{3/4} = 25^{0.75} = 11.18$.

$N_S = 1450 \times 0.2828/11.18 = \mathbf{36.68}$.

Exam tip: Do not mix the pump formula (Q, $H^{3/4}$) with the turbine formula (P, $H^{5/4}$). Low N_S → radial-flow pump, high N_S → axial-flow pump.

What next?

Day 82 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 83/90: Full-Length Mock 5. Comment **GATE90** on our reel to get all links.
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